Suppose ω_t satisfy

$$Ric(\omega_t) = t\omega_t + (1-t)[D]$$

for $t \in (0, T)$, $T < 1$.

As before we can take GH limit

$$(M, \omega_{t_k}) \longrightarrow (Y, d_Y)$$

and we have embeddings

$$\Phi_k : M \hookrightarrow \mathbb{CP}^N$$

using sections of K_M^{-m}, converging to an embedding $\Phi : Y \hookrightarrow \mathbb{CP}^N$

$W = \Phi(Y)$ is a normal variety

Can assume that

$$\Phi_k(D) \longrightarrow D_\infty \subset W$$

i.e. the pair (W, D_∞) is in closure of the $SL_{\mathbb{C}}(N+1)$-orbit of $(\Phi_1(M), \Phi_1(D))$

W admits a (singular) solution ω of

$$(*) \quad Ric(\omega) = T\omega + (1-T)[D_\infty]$$

This eq^n can be interpreted as eq^n for metrics on K_W^{-1} (a $\mathbb{Q}$-line bundle)

D_∞ = zero set of a section s_0 of K_W^{-1} (or $K_{W_{reg}}^{-1}$). Define singular metric h_{D_∞} on K_W^{-1} so that $|s_0|^2_{h_{D_\infty}} = 1$

i.e. if in a local trivialization s for K_W^{-1}

$s_0 = z s$, then

$$|s|^2_{h_{D_\infty}} = \frac{1}{|z|^2} = e^{-\psi}$$

for $\psi = \ln |z|^2$

(*) is equivalent to having a metric $h = e^{-\varphi}$ with bounded potential on K_W^{-1} s.t.

$$\omega_\varphi^n = e^{-T\varphi - (1-T)\psi} \text{ as metrics on } K_W^{-1}$$

As before, this has 2 consequences:

1) $\mathrm{Aut}(W, D_\infty)$ is reductive

2) Twisted Futaki invariant $DF(W, (1-T)D_\infty, \lambda) = 0$ for all $\lambda : \mathbb{C}^* \hookrightarrow \mathrm{Aut}(W, D_\infty)$

$$DF(W, (1-T)D_\infty, \lambda) = T \cdot DF(W, \lambda) + (1-T) \cdot \frac{n}{V} \int_W \theta_\lambda \left(m[D_\infty] - \omega \right) \wedge \omega^{n-1}$$

1) $\Longrightarrow$ we can find $\lambda : \mathbb{C}^* \hookrightarrow SL(N+1)$

s.t. $(W, D_\infty) = \lim_{t \to 0} \lambda(t) \cdot (M, D)$

for a suitable embedding of $(M, D) \hookrightarrow \mathbb{CP}^N$

2) $\Longrightarrow$ $DF(M, (1-T)D, \lambda) = 0$.

$f(t) = DF(M, (1-t)D, \lambda)$ is linear in t

and one can show that

$f(t) \geq 0$ for $t < T$

since we can solve our eq^n for $t < T$.

and $f(t) = 0$ only if $(W, D_\infty) \simeq (M, D)$
(for $t < T$)

$\Longrightarrow$ Either $f(1) < 0$, i.e. $DF(M, \lambda) < 0$

so M is not K-stable

Or $f(t) = 0$ $\forall t$, so $(W, D_\infty) \simeq (M, D)$

So on M we have soln. ω of

$$Ric(\omega) = T\omega + (1-T)[D]$$

Need to show that ω has correct conical singularities along D.

Then we can keep going along continuity method.

$\square$

<u>Example</u> Easier to work with Ricci flat Kähler cone metrics rather than projective KE metrics. Alternatively the link of a Ricci flat Kähler cone metric is a "Sasaki–Einstein" mfld

$$\left(\omega_{FS} \text{ of } \mathbb{CP}^n, \ g_{round} \text{ on } S^{2n+1}, \ \omega_{Euc} \text{ on } \mathbb{C}^{n+1} \right)$$

<u>Theorem</u> $X_{p,q} = \{ uv + z^p + w^q = 0 \} \subset \mathbb{C}^4$

admits a Ricci flat Kähler cone metric

$$\iff 2p > q \ \& \ 2q > p.$$

If e.g. $2q \leq p$, consider $\lambda \hookrightarrow GL(4, \mathbb{C})$

diagonal : $\lambda(t)(u, v, z, w) = (u, v, tz, w)$

$$\lambda(t) \cdot X_{p,q} \longrightarrow Y = \{ uv + w^q = 0 \} \subset \mathbb{C}^4$$

normal variety

$$\{ uv + (tz)^p + w^q = 0 \}$$

$$DF(Y, \lambda) = \underset{c>0}{c} \cdot (2q - p) \leq 0$$